ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Protista	Euglena.
2	**(b)** gills throughout life	Fish breathe with gills.
3	**(d)** Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris	Hierarchy.
4	**(a)** monocot	Monocot.
5	**(b)** fungus	Saprophyte.
6	**(c)** bird	Aves.
7	**(b)** more closely related than species in the same family but different genera	Closer relation.
8	**(c)** monocot seed	One cotyledon.
9	**(c)** chordate	Chordata.
10	**(a)** platypus	Monotreme.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Protista (1); Unicellular eukaryote (1).	1+1
12	Breathes by lungs, has mammary glands (1); Warm-blooded and gives birth (1).	1+1
13	Roundworm: cylindrical, unsegmented (1); Earthworm: segmented body (1).	1+1
14	They are non-cellular (1); They show life only inside host cells (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Banyan: dicot angiosperm; rice: monocot (1); Fern: pteridophyte; moss: bryophyte (1); Spirogyra: alga; pine: gymnosperm (1).	1+1+1
16	Aves (1); Hollow bones for flight (1); Warm-blooded (1).	1+1+1
17	Flat, bilaterally symmetrical (1); Tapeworm and liver fluke (1); Cause diseases (1).	1+1+1
	OR: Earthworm: Annelida; hydra: Cnidaria (1); Octopus: Mollusca; tapeworm: Platyhelminthes (1); Starfish: Echinodermata; cockroach: Arthropoda (1).	1+1+1
18	Group of organisms that can breed and produce fertile offspring (1); Horse and donkey produce sterile mule (1); So they are different species (1).	1+1+1
	OR: Simple: Porifera (1); More complex: Cnidaria, worms (1); Chordates with the highest complexity (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mushroom: Fungi; bacterium: Monera (1); Euglena: Protista (1); Rose: angiosperm; fern: pteridophyte; moss: bryophyte (1); Whale: mammal; snake: reptile; pigeon: bird (1); Earthworm: annelid (1).	1+1+1+1+1
	OR: Birds: feathers, hollow bones, wings (1); Streamlined body (1); Fish: fins, gills (1); Swim bladder, scales (1); Streamlined (1).	1+1+1+1+1
20	Pisces (1); Amphibia (1); Reptilia (1); Aves (1); Mammalia (1).	1+1+1+1+1
	OR: Food and medicines (1); Ecosystem stability (1); Threats: deforestation, pollution (1); Protected areas (1); Laws and awareness (1).	1+1+1+1+1

Mistakes to avoid:

- Writing scientific names wrongly: *homo Sapiens* or *Homo Sapiens*; the genus has a capital letter, the species a small letter, and both are italicised or underlined separately.
- Treating the kingdom as the smallest group and the species as the largest; the hierarchy runs from kingdom (largest) to species (smallest).
- Putting fungi (mushroom, yeast) in the plant kingdom; they lack chlorophyll and have chitin in their cell wall.

- Mixing up gymnosperms (naked seeds) and angiosperms (seeds enclosed in fruits), or monocots and dicots (one or two cotyledons).
- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.
- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** kingdom	Highest rank.
2	**(c)** pea	Two cotyledons.
3	**(a)** fern	Pteridophyta.
4	**(c)** bat	Mammal.
5	**(c)** species	Lowest rank.
6	**(a)** maize	Parallel venation.
7	**(a)** alga	Thallophyta.
8	**(b)** whale	Breathes air.
9	**(d)** Mangifera indica	Capital for genus only.
10	**(b)** sexual reproduction	Reproduction.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cold-blooded: body temperature varies, e.g. frog (1); Warm-blooded: constant, e.g. human (1).	1+1
12	Rod-like structure along the back (1); Chordates at some stage (1).	1+1
13	Bryophytes lack vascular tissue (1); Pteridophytes have true roots, stems and vascular tissue (1).	1+1
14	Universal language (1); Avoids confusion of local names (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell wall: present in plants, absent in animals (1); Nutrition: autotrophic vs heterotrophic (1); Movement and growth: limited and localised vs mobile and diffuse (1).	1+1+1
16	Algae have chlorophyll; fungi don't (1); Algae autotrophic; fungi heterotrophic (1); Cell wall: cellulose vs chitin (1).	1+1+1
17	Naked seeds and cones (1); Ancient group with few species (1); Little change over millions of years (1).	1+1+1
	OR: Bacteria, fungi and protists do not fit plants or animals (1); Cell type and nutrition considered (1); More natural grouping (1).	1+1+1
18	Uses: yeast, antibiotics, mushrooms (1); Harm: spoilage of food (1); Diseases of plants and humans (1).	1+1+1
	OR: Easy identification (1); Study relationships (1); Organise data about millions of species (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Aristotle's classification (1); Two kingdoms (1); Five kingdoms (1); Three domains with molecular evidence (1); New discoveries change classification (1).	1+1+1+1+1
	OR: Genus capital, species small (1); Italics or underline (1); Homo sapiens (man) (1); Panthera tigris (tiger), Oryza sativa (rice) (1); Mangifera indica (mango) (1).	1+1+1+1+1
20	Nutrition (1); Cell wall (1); Reproduction by spores (1); Habitat (1); Examples (1).	1+1+1+1+1
	OR: Eggs laid in water (1); Tadpole with gills (1); Adult with lungs and moist skin (1); Metamorphosis (1); Dependence on water (1).	1+1+1+1+1

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- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.

- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** pine	Gymnosperm.
2	**(c)** crocodile	Reptilia.
3	**(d)** Panthera leo	Genus capital, species small.
4	**(a)** fruits	Naked.
5	**(c)** dicot	Dicot.
6	**(d)** Reptilia	Dry scaly skin.
7	**(b)** order	Higher ranks shared.
8	**(d)** spores	Pteridophytes.
9	**(b)** insect	Class Insecta.
10	**(a)** frog	Poikilothermic.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dry scaly skin prevents water loss (1); Eggs have shells (1).	1+1
12	It has hair and mammary glands (1); Gives birth to young ones, no feathers (1).	1+1
13	Backbone present vs absent (1); Endoskeleton vs various (1).	1+1
14	Yeast and mushroom: Fungi (1); Amoeba: Protista; bacterium: Monera (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Skin covering (1); Respiratory organs (1); Heart chambers (1).	1+1+1
16	Insects: three pairs of legs, wings (1); Arachnids: four pairs of legs, no wings (1); Body parts: three vs two (1).	1+1+1
17	Both reproduce by spores (1); Mosses: no vascular tissue (1); Ferns: vascular tissue and true roots (1).	1+1+1
	OR: Amphibia (1); Three-chambered heart (1); Moist skin and aquatic eggs (1).	1+1+1
18	Food for many animals and humans (1); Pearl and shell (1); Starfish control populations (1).	1+1+1
	OR: Monera: prokaryotic; various nutrition (1); Protista and Fungi: eukaryotic; mixed (1); Plantae autotrophic; Animalia heterotrophic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Features of each (1); Symmetry (1); Coelom (1); Segmentation (1); Examples (1).	1+1+1+1+1
	OR: Hair, mammary glands (1); Four-chambered heart, diaphragm (1); Live birth (1); Platypus lays eggs (1); Bat flies; whale lives in water (1).	1+1+1+1+1
20	Algae: no vascular tissue (1); Bryophytes: no vascular tissue (1); Pteridophytes: vascular tissue, spores (1); Gymnosperms: naked seeds (1); Angiosperms: flowers and fruits (1).	1+1+1+1+1
	OR: Bacteria: cholera (1); Protozoa: malaria (1); Fungi: ringworm (1); Viruses: influenza (1); Algae: useful as food (1).	1+1+1+1+1

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